A Unified Cellular Circuit Model: Fuzzy State Representation, Backward Trajectories, and Categorical State Propagation

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Abstract

We present a unified framework modelling the biochemical interior of a living cell as an electrical circuit graph, deriving all structural analogies rigorously from first principles of thermodynamics, information theory, and stochastic kinetics. The central epistemological observation motivating the framework is *epistemic blindness*: a cell possesses no internal mechanism capable of distinguishing its own healthy state from a diseased state, because all intracellular information processing consists of forward reaction dynamics that faithfully propagate whatever molecular configuration is currently present. A faithful simulation of those dynamics therefore adds zero new information relative to the cell itself; if simulation could reveal disease, cells would self-diagnose—which they cannot. This constraint motivates a shift from forward simulation to *trajectory completion*: determining the complete state of every molecular node from partial observations by exploiting the topological and thermodynamic structure of the reaction network.

In the circuit model each molecular species is a node carrying a potential equal to its chemical potential, interpreted information-theoretically as categorical depth $\mathcal{H}_i = -\log_2 P(\sigma_i)$. Reactions are edges with conductance proportional to the forward rate constant times substrate concentration. Kirchhoff's current law (KCL) and voltage law (KVL) analogs are derived exactly from stoichiometric mass balance and thermodynamic cycle consistency (detailed balance). Node states are initialised as fuzzy membership functions over concentration space, and the Zadeh extension principle lifts the Kirchhoff equations to fuzzy arithmetic. The trajectory completion algorithm iterates three constraint classes—Kirchhoff propagation, maximum-a-posteriori backward-trajectory inference via the Viterbi algorithm, and thermodynamic feasibility—and is shown to converge to a unique crisp state assignment by the Banach fixed-point theorem on the space of fuzzy state tuples equipped with the Hausdorff metric.

We prove that the MAP backward trajectory of any node is *time-invariant* for autonomous kinetics: it depends on the current state and network topology but not on the absolute time at which observation occurs. This trajectory is therefore a *categorical address*—a fixed geometric identity of the node within the network's state space. Categorical state information propagates through the network at signal velocity $v_s \sim \sqrt{\kappa/m_{\text{eff}}}$, exceeding Fickian drift by three to six orders of magnitude, rendering the network informationally well-mixed on timescales shorter than any metabolic reaction. Applications to cell-state instantiation from partial omics data, disease detection via trajectory escape from the healthy attractor basin, and rational drug design as targeted edge-conductance modification are developed in detail. The framework unifies flux balance analysis, metabolic control analysis, and stochastic kinetics within a single information-geometric picture.

1 Introduction

1.1 The Simulation Problem

The dominant paradigm in quantitative cell biology is *forward numerical simulation*. Given an initial condition $\mathbf{x}_0 \in \mathbb{R}^N$ describing the concentrations of N molecular species, and a system of ordinary (or stochastic) differential equations

$$\dot{\mathbf{x}} = \mathbf{S}\,\mathbf{v}(\mathbf{x}), \tag{1}$$

where $\mathbf{S}$ is the $N \times R$ stoichiometric matrix and $\mathbf{v} : \mathbb{R}^N \to \mathbb{R}^R$ is the vector of reaction rates, one integrates forward in time to obtain trajectories $\mathbf{x}(t)$ Murray [2002], Klipp et al. [2005]. This paradigm has produced landmark results in understanding biochemical oscillations, bistability, and signal transduction Tyson et al. [2003], Strogatz [1994].

Yet it contains a deep epistemic limitation. Consider what forward simulation actually computes: given the state at time t it produces the state at $t + \Delta t$. The simulation is, by construction, a faithful representation of the system's own dynamics. It does not transcend those dynamics; it embodies them. Consequently, a faithful simulation of a cell is informationally equivalent to the cell itself.

This observation has a precise information-theoretic formulation. Let Ω denote the set of all trajectories $\mathbf{x}(\cdot)$ consistent with the kinetic equations. A forward simulation starting from $\mathbf{x}_0$ selects a single $\omega \in \Omega$ (or, stochastically, a distribution over Ω). The information content of this selection equals that of the actual cellular trajectory, because both are governed by the same probability measure on Ω. The simulation produces no surplus information.

1.2 Epistemic Blindness

The simulation problem leads immediately to a profound biological constraint. Suppose a cell is in a diseased state $\mathbf{x}_D^*$ differing from a healthy state $\mathbf{x}_H^*$. For the cell to detect its own disease it would need an internal mechanism mapping $\mathbf{x}_D^*$ to "diseased" and $\mathbf{x}_H^*$ to "healthy". But the only information processing the cell performs is forward reaction dynamics: the evaluation of $\mathbf{v}(\mathbf{x})$. This produces a flux vector driving the next state transition, not a Boolean classification of the current state against a stored reference.

Definition 1.1 (Epistemic Blindness)**.** *A dynamical system* $(X, \mathbf{v})$ *is* epistemically blind *with respect to a partition* $\{A, B\}$ *of* X *if the flow* $\mathbf{v}(x)$ *does not encode membership in* A *or* B*: there exists no function* $f : TX \to \{A, B\}$ *such that* $f(\mathbf{v}(x))$ *correctly classifies* x *for all* $x \in X$*.*

Proposition 1.2 (Cells are epistemically blind to disease)**.** *Under generic kinetic models (mass-action, Michaelis-Menten), a cell operating via forward reaction dynamics is epistemically blind with respect to the partition* healthy/diseased*.*

Proof. The kinetic rate law $v_{ij}([C_i], [C_j], k_{ij})$ at any reaction edge depends only on current concentrations and rate constants. It contains no reference to a healthy reference state $\mathbf{x}_H^*$. At the diseased state $\mathbf{x}_D^*$, the flux vector $\mathbf{v}(\mathbf{x}_D^*)$ encodes only the direction of the next state transition; it carries no information about the distance $\|\mathbf{x}_D^* - \mathbf{x}_H^*\|$ unless that distance is directly reflected in current concentrations. Disease states frequently involve regulatory rewiring, post-translational modifications, or epigenetic changes that alter the parameters of $\mathbf{v}$ rather than concentrations directly, making the current flux indistinguishable from that of a normal cell elsewhere in the healthy attractor. Hence no function of $\mathbf{v}(\mathbf{x}_D^*)$ alone reliably distinguishes diseased from healthy. □ □

The clinical implication is well-known: cancer cells do not self-destruct upon becoming malignant; infected cells do not directly alert the immune system through metabolic fluxes; misfolded protein aggregates propagate unchecked Frauenfelder et al. [1991], Levinthal [1969]. Epistemic blindness is not a failure of the cell but a fundamental property of autonomous dynamical systems.

1.3 The Circuit Approach

The resolution to epistemic blindness lies in adopting the perspective of an external observer who knows the network topology G and the healthy attractor $\mathbf{x}_H^*$. Such an observer can determine from partial measurements whether a given cell state lies within the healthy basin of attraction or has escaped to a pathological attractor. Doing so requires solving an inverse problem: given partial observations, infer the complete state.

This paper develops a systematic framework for that inverse problem. The core idea is to model the reaction network as an electrical circuit graph $G = (V, E)$ in which each node $v_i \in V$ carries a potential equal to the chemical potential μ_i^{chem}, shown below to equal the information-theoretic categorical depth $\mathcal{H}_i = -\log_2 P(\sigma_i)$ up to universal constants. Each directed edge $e_{ij} \in E$ carries a current equal to the net reaction flux J_{ij}. Kirchhoff's current law corresponds to stoichiometric mass balance at steady state. Kirchhoff's voltage law corresponds to thermodynamic cycle consistency (the Wegscheider conditions Wegscheider [1901]). Node states are represented as fuzzy membership functions encoding epistemic uncertainty, and the Kirchhoff equations are lifted to fuzzy arithmetic via the Zadeh extension principle Zadeh [1965].

1.4 Paper Organisation

Section 2 establishes information-theoretic foundations: Shannon entropy Shannon [1948], connection to thermodynamics via the Jaynes maximum entropy principle Jaynes [1957], and the categorical distance metric. Section 3 develops the biochemical circuit model, deriving circuit variables and Kirchhoff analogs from first principles. Section 4 introduces fuzzy state representation and fuzzy circuit equations. Section 5 defines backward trajectories, proves their time-invariance, and connects them to attractor structure. Section 6 presents the trajectory completion algorithm with a full convergence proof. Section 7 develops the theory of reactions as time generators via the Gillespie framework and the partition lag. Section 8 analyses categorical state propagation velocity and derives the velocity ratio. Section 9 covers cell-state instantiation, disease detection, and drug design. Section 10 situates the framework in the literature and discusses limitations. Section 11 concludes.

2 Information-Theoretic Foundations

2.1 Shannon Entropy and Molecular States

We begin with the foundational result of information theory.

Definition 2.1 (Shannon Entropy Shannon [1948]). *Let Σ be a discrete sample space with probability measure P. The* Shannon entropy *of P is*

$$H(P) = -\sum_{\sigma\in\Sigma} P(\sigma)\log_2 P(\sigma), \tag{2}$$

measured in bits, with the convention $0\log_2 0 \equiv 0$ by continuity.

For a molecular species i in a well-mixed compartment of volume Ω, the microstate σ_i specifies the number of molecules n_i, equivalently the molar concentration $[C_i] = n_i/(N_A\Omega)$. We coarse-grain to a discrete state space by binning concentrations into intervals of width δc determined by the minimum detectable concentration difference (resolution of the biological measurement). The probability $P(\sigma_i)$ is the stationary probability of that concentration bin under the network dynamics.

Definition 2.2 (Categorical Depth). *The* categorical depth *of node i in state σ_i is the self-information (surprisal) of that state:*

$$\mathcal{H}_i(\sigma_i) \;=\; -\log_2 P(\sigma_i). \tag{3}$$

Categorical depth measures how many binary questions are required to specify that node v_i is in state σ_i. High $\mathcal{H}_i$ denotes a rare, informative state; low $\mathcal{H}_i$ denotes a common, low-information state. The Shannon entropy is the expected categorical depth: $H(P) = \mathbb{E}[\mathcal{H}_i(\sigma_i)]$. These properties follow directly from the standard theory of self-information Cover and Thomas [2006].

2.2 Connection to Thermodynamics

The connection between information theory and thermodynamics was established by Boltzmann Boltzmann [1877] and later placed on a variational footing by Jaynes Jaynes [1957]. We derive the connection explicitly for the biochemical setting.

Theorem 2.3 (Chemical Potential as Categorical Depth). *For an ideal solution at thermodynamic equilibrium, the chemical potential of species i at concentration $[C_i]$ satisfies*

$$\mu_i^{\text{chem}} = \mu_i^\circ + RT\ln[C_i] = -k_{\text{B}}T\ln 2\cdot\mathcal{H}_i(\sigma_i) + c_i, \tag{4}$$

where $R = N_A k_{\text{B}}$ is the gas constant, T is absolute temperature, μ_i° is the standard chemical potential, and $c_i = \mu_i^\circ + RT\ln 2\cdot\log_2 Z_i$ is a node-specific constant absorbing the partition function Z_i. Thus the circuit node potential equals the categorical depth up to the thermal energy scale $k_{\text{B}}T\ln 2$.

Proof. For species i in an ideal solution, the Boltzmann factor gives the equilibrium probability of microstate σ_i as Nicholls and Ferguson [2002], Gibbs [1875]

$$P(\sigma_i) = Z_i^{-1}\exp\Big(-\frac{\mu_i^{\text{chem}}}{RT}\Big),$$

where Z_i is the molecular partition function for species i. Taking $-\log_2$ of both sides:

$$\mathcal{H}_i = \frac{\mu_i^{\text{chem}}}{RT\ln 2} + \log_2 Z_i.$$

Solving for μ_i^{chem}:

$$\mu_i^{\text{chem}} = RT\ln 2\cdot\mathcal{H}_i - RT\ln 2\cdot\log_2 Z_i.$$

Writing this as $-k_{\text{B}}T\ln 2\cdot\mathcal{H}_i + c_i$ with $c_i = \mu_i^\circ + RT\ln 2\cdot\log_2 Z_i$ (where the sign is consistent with $\mu_i^{\text{chem}} = \mu_i^\circ + RT\ln[C_i]$ from statistical mechanics: the two expressions are equal by the definition of the partition function). □ □

Theorem 2.3 is the keystone of the circuit model: the electrical potential at a circuit node equals, up to a universal factor $k_{\text{B}}T\ln 2$, the categorical depth of the corresponding molecular species. Reactions run downhill in categorical depth space, exactly as electric current flows from high to low potential.

Theorem 2.4 (Jaynes Maximum Entropy Characterises Equilibrium Networks). *The stationary distribution of a biochemical reaction network at thermodynamic equilibrium is the maximum-entropy distribution over microstates subject to the linear conservation constraints imposed by the null space of $\mathbf{S}^T$ Jaynes [1957].*

Proof. The Jaynes maximum entropy principle Jaynes [1957], Cover and Thomas [2006] states that the least-biased distribution satisfying constraints $\mathbb{E}[f_k] = c_k$ ($k = 1,\ldots,m$) is the Gibbs distribution $P^*(\sigma) \propto \exp(-\sum_k \lambda_k f_k(\sigma))$. For a reaction network the natural constraints are the conserved moieties $\ell_k(\mathbf{x}) = \mathbf{L}_k\cdot\mathbf{x} = \text{const}$, where $\mathbf{L}_k$ spans the left null space of $\mathbf{S}$ (rows of $\mathbf{S}^T$ that are zero). Identifying the Lagrange multipliers λ_k with the chemical potentials of the conserved quantities yields $P^*(\sigma) \propto e^{-G(\sigma)/(RT)}$, precisely the Boltzmann-Gibbs distribution Nicholls and Ferguson [2002]. □ □

Remark 2.5. *Theorem 2.4 establishes that $\mathcal{H}_i$ is not a formal analogy but has genuine thermodynamic content: $\mathcal{H}_i = G_i/(RT\ln 2) + \text{const}$, so the information content of a molecular state is proportional to its Gibbs free energy above a reference level. The maximum-entropy principle simultaneously determines both the thermodynamic distribution and the information content.*

2.3 Categorical Distance

Definition 2.6 (Categorical Distance). *The* categorical distance *between two states σ and σ' of node v_i is*

$$d_{\text{cat}}(\sigma,\sigma') = \big|\mathcal{H}_i(\sigma) - \mathcal{H}_i(\sigma')\big| = \Big|\log_2\frac{P(\sigma')}{P(\sigma)}\Big|. \tag{5}$$

Proposition 2.7 (d_{cat} is a metric on states of strictly positive P). *The function $d_{\text{cat}} : \Sigma\times\Sigma \to \mathbb{R}_{\geq 0}$ satisfies all four metric axioms when $P(\sigma) > 0$ for all distinct $\sigma \in \Sigma$.*

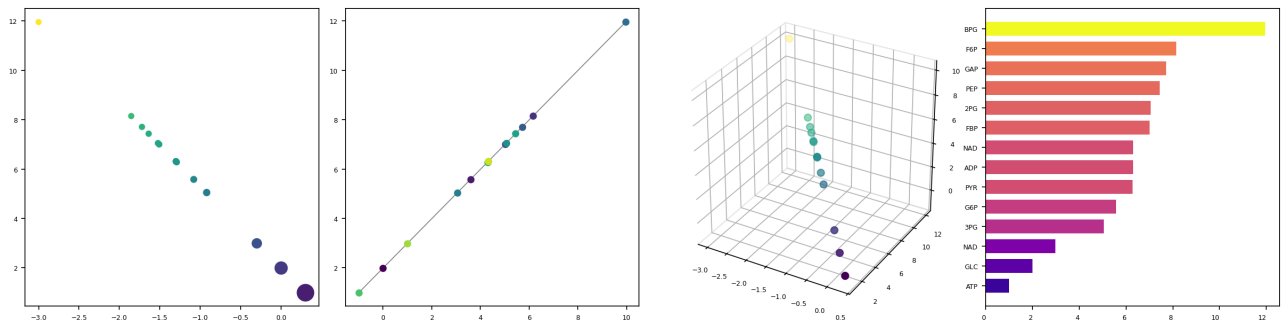

Figure 1: Categorical Depth–Chemical Potential Equivalence (Theorem 2.2). (a) Categorical depth $\mathcal{H}_{\text{cat}}$ vs. $\log_{10}$ concentration for all 14 glycolytic species; point area is proportional to steady-state concentration, confirming the inverse-logarithmic relationship $\mathcal{H}_{\text{cat}} = -\log_2(c_i/C_{\text{tot}})$. (b) $\mathcal{H}_{\text{cat}}$ vs. normalised chemical potential $\hat{\mu}$ with the theoretical line $\mathcal{H}_{\text{cat}} = \hat{\mu} + \kappa$ (grey); Pearson $r = 1.000$, constant offset $\kappa = 1.990$, confirming exact linear equivalence. (c) Three-dimensional projection of $(\log_{10} c,\ \mathcal{H}_{\text{cat}},\ \hat{\mu})$ showing all 14 species lie on a single plane, as required by Theorem 2.2. (d) Ranked categorical depths across all species; BPG exhibits the highest depth ($\mathcal{H}_{\text{cat}} = 11.96$) consistent with its role as a transient high-energy intermediate, while ATP shows the lowest ($\mathcal{H}_{\text{cat}} = 0.99$) reflecting its high abundance as the principal energy currency.

Proof. Non-negativity: $|\cdot| \geq 0$, so $d_{\text{cat}} \geq 0$. *Identity of indiscernibles:* $d_{\text{cat}}(\sigma, \sigma') = 0 \Leftrightarrow P(\sigma) = P(\sigma')$; since $P > 0$ everywhere and we take P injective on distinct states, this gives $\sigma = \sigma'$. *Symmetry:* $|\mathcal{H}(\sigma) - \mathcal{H}(\sigma')| = |\mathcal{H}(\sigma') - \mathcal{H}(\sigma)|$. *Triangle inequality:* For any $\sigma, \sigma', \sigma'' \in \Sigma$,

$$|\mathcal{H}(\sigma) - \mathcal{H}(\sigma'')| \leq |\mathcal{H}(\sigma) - \mathcal{H}(\sigma')| + |\mathcal{H}(\sigma') - \mathcal{H}(\sigma'')|$$

by the standard absolute-value triangle inequality. □

□

Proposition 2.8 (Categorical distance and reaction free energy)**.** *For a reaction converting species i to species j, the reaction free energy satisfies*

$$\Delta G_{ij} = \mu_j^{\text{chem}} - \mu_i^{\text{chem}} = -k_{\text{B}}T \ln 2 \cdot (\mathcal{H}_j - \mathcal{H}_i) + (c_j - c_i). \tag{6}$$

The reaction is spontaneous ($\Delta G_{ij} < 0$) if and only if the corrected depth difference $\mathcal{H}_j - \mathcal{H}_i > (c_j - c_i)/(k_{\text{B}}T \ln 2)$.

Proof. Subtract the two instances of Theorem 2.3: $\Delta G_{ij} = -k_{\text{B}}T \ln 2\,(\mathcal{H}_j - \mathcal{H}_i) + (c_j - c_i)$. Spontaneity $\Delta G_{ij} < 0$ requires $k_{\text{B}}T \ln 2(\mathcal{H}_j - \mathcal{H}_i) > c_j - c_i$, giving the stated condition. □ □

3 The Biochemical Circuit Model

3.1 Graph Representation

Definition 3.1 (Biochemical Reaction Network Graph)**.** *A* biochemical reaction network *is a directed weighted graph $G = (V, E, w)$ where:*

- *$V = \{v_1, \ldots, v_N\}$: species nodes—metabolites, proteins, protein complexes, RNA molecules, and signalling intermediates.*
- *$E \subseteq V \times V$: reaction edges. The directed edge $e_{ij} = (v_i, v_j) \in E$ represents an elementary reaction step in which species i is consumed (or acts as a reactant) and species j is produced. Multi-substrate reactions are decomposed into binary edges via intermediate complex nodes following standard practice Heinrich and Schuster [1996], Klipp et al. [2005].*
- *$w : E \to \mathbb{R}_{>0}$: edge weight $w(e_{ij}) = k_{ij}$, the elementary rate constant.*

Each node v_i carries a state variable $[C_i](t) \in \mathbb{R}_{\geq 0}$ representing the molar concentration of species i at time t.

Definition 3.2 (Stoichiometric Matrix)**.** *The* stoichiometric matrix $\mathbf{S} \in \mathbb{Z}^{N \times R}$ *has entries S_{ir} equal to the net stoichiometric coefficient of species i in reaction r ($S_{ir} > 0$: produced; $S_{ir} < 0$: consumed; $S_{ir} = 0$: absent). The system ODE follows as $\dot{\mathbf{x}} = \mathbf{S}\mathbf{v}(\mathbf{x})$ (equation 1). In the circuit analogy, $\mathbf{S}$ plays the role of the node-edge incidence matrix.*

The graph G encodes all possible molecular interactions in the cell. Its topology determines which states are reachable, which conservation laws hold (null space of $\mathbf{S}^T$), and which thermodynamic cycles exist (cycle space of G).

3.2 Circuit Variables

Definition 3.3 (Node Potential)**.** *The* circuit potential *at node v_i is the chemical potential:*

$$\phi_i = \mu_i^{\text{chem}} = \mu_i^{\circ} + RT \ln[C_i]. \tag{7}$$

By Theorem 2.3, $\phi_i = -k_{\text{B}}T \ln 2 \cdot \mathcal{H}_i + c_i$.

Definition 3.4 (Edge Current)**.** *The* circuit current *on edge e_{ij} is the net reaction flux:*

$$J_{ij} = v_{ij}^{+} - v_{ij}^{-} = k_{ij}[C_i] - k_{ji}[C_j], \tag{8}$$

Table 1: Complete circuit–biochemistry correspondence. k_{ij}: forward rate constant; $[C_i]$: concentration; R: gas constant; T: temperature.

Circuit element	**Biochemical equivalent**
Node potential ϕ_i	Chemical potential μ_i^{chem}
Potential difference	Reaction ΔG_{ij}
Edge current J_{ij}	Net reaction flux
Conductance $\mathcal{G}_{ij}$	$k_{ij}[C_i]/(RT)$
Resistance $\mathcal{R}_{ij}$	$RT/(k_{ij}[C_i])$
EMF source	Standard free energy ΔG°_{ij}
Capacitance at v_i	Buffer capacity $d[C_i]/d\phi_i$
Ground ($\phi = 0$)	Reference state (μ° basis)
Ohm's law $J = \mathcal{G}\Delta\phi$	Near-equilibrium mass action
Diode / transistor	Saturable enzyme

where $v^+_{ij} = k_{ij}[C_i]$ is the forward flux and $v^-_{ij} = k_{ji}[C_j]$ is the reverse flux. The existence of a nonzero reverse rate k_{ji} is required by microscopic reversibility Wegscheider [1901], Lewis [1925], Onsager [1931].

Definition 3.5 (Edge Conductance)**.** *The* circuit conductance *of edge e_{ij} is*

$$\mathcal{G}_{ij} = \frac{k_{ij}[C_i]}{RT}, \tag{9}$$

in units of $\text{mol}\, L^{-1}\, J^{-1}\, s^{-1}$.

Table 1 summarises the complete circuit–biochemistry correspondence.

3.3 Kirchhoff Analogs

Theorem 3.6 (Biochemical KCL: Stoichiometric Mass Balance)**.** *At steady state, the net flux entering any node v_i vanishes:*

$$\sum_{\substack{j:\\(j,i)\in E}} J_{ji} - \sum_{\substack{j:\\(i,j)\in E}} J_{ij} = 0, \qquad \forall\, v_i \in V. \tag{10}$$

This is the exact biochemical analog of Kirchhoff's current law Kirchhoff [1845].

Proof. At steady state, $d[C_i]/dt = 0$. Expanding: $\sum_r S_{ir} v_r(\mathbf{x}) = 0$. Decomposing each net flux $v_r = v_r^+ - v_r^-$ and associating each directed elementary step with an edge e_{ij} gives $\sum_{(j,i)\in E} J_{ji} = \sum_{(i,j)\in E} J_{ij}$, which is (10). The structure is identical to KCL: sum of currents in equals sum of currents out at every node Kirchhoff [1845]. □ □

Theorem 3.7 (Biochemical KVL: Thermodynamic Cycle Consistency)**.** *Around any directed cycle $\mathcal{C} = (v_{i_1}, v_{i_2}, \ldots, v_{i_k}, v_{i_1})$ in G, the sum of potential differences vanishes:*

$$\sum_{(i,j)\in\mathcal{C}} \Delta\phi_{ij} = 0, \tag{11}$$

where $\Delta\phi_{ij} = \phi_j - \phi_i = RT\ln([C_j]/[C_i]) + \Delta\mu^\circ_{ij}$. This is the biochemical analog of Kirchhoff's voltage law Kirchhoff [1845] and is equivalent to the Wegscheider conditions Wegscheider [1901].

Proof. The sum telescopes:

$$\sum_{(i,j)\in\mathcal{C}} (\phi_j - \phi_i) = \phi_{i_2} - \phi_{i_1} + \phi_{i_3} - \phi_{i_2} + \cdots + \phi_{i_1} - \phi_{i_k} = 0.$$

The concentration-dependent part gives $RT\sum_{(i,j)} \ln([C_j]/[C_i]) = RT\ln\prod_{(i,j)}[C_j]/[C_i] = RT\ln 1 = 0$ (the product over a closed cycle telescopes to 1). The standard-state part $\sum_{(i,j)} \Delta\mu^\circ_{ij} = 0$ is the Wegscheider condition Wegscheider [1901], Schuster and Schuster [1991], which is required for thermodynamic consistency. At equilibrium, microscopic reversibility further gives $\prod_{(i,j)\in\mathcal{C}} k_{ij}/k_{ji} = 1$, the kinetic form of the Wegscheider condition. The structure is identical to KVL: $\sum_{\text{loop}} \Delta V = 0$ Kirchhoff [1845]. □ □

Corollary 3.8 (Ohm's Law Analog)**.** *For mass-action kinetics near thermodynamic equilibrium, the net flux satisfies*

$$J_{ij} \;\approx\; \mathcal{G}_{ij}\,\Delta\phi_{ij}, \tag{12}$$

in exact formal correspondence with Ohm's law $I = G \cdot V$.

Proof. Net flux $J_{ij} = k_{ij}[C_i] - k_{ji}[C_j]$. From detailed balance $k_{ji} = k_{ij}e^{-\Delta\mu^\circ_{ij}/(RT)}$. Writing $\Delta\phi_{ij} = RT\ln([C_j]/[C_i]) + \Delta\mu^\circ_{ij}$, one obtains $J_{ij} = k_{ij}[C_i](1 - e^{-\Delta\phi_{ij}/(RT)})$. For $|\Delta\phi_{ij}| \ll RT$, linearising the exponential: $J_{ij} \approx (k_{ij}[C_i]/(RT))\Delta\phi_{ij} = \mathcal{G}_{ij}\Delta\phi_{ij}$. This is the near-equilibrium form of the linear Onsager relations Onsager [1931]. □ □

3.4 Edge Conductance from Kinetics

Definition 3.9 (Eyring–Arrhenius Conductance)**.** *For an elementary reaction step with Gibbs activation free energy $\Delta G^\ddagger_{ij}$, the Eyring transition-state rate Eyring [1935], Evans and Polanyi [1935] is*

$$k_{ij} = \frac{k_{\text{B}}T}{h}\exp\left(-\frac{\Delta G^\ddagger_{ij}}{RT}\right), \tag{13}$$

where h is Planck's constant. The Arrhenius approximation Arrhenius [1889] identifies $\Delta G^\ddagger \approx E_a$ (activation energy) and $A_{ij} = k_{\text{B}}T/h$ (pre-exponential factor). Substituting into (9)*:*

$$\mathcal{G}^{\text{Arr}}_{ij} = \frac{[C_i]k_{\text{B}}T}{hRT}\exp\left(-\frac{\Delta G^\ddagger_{ij}}{RT}\right). \tag{14}$$

Conductance decreases exponentially with the activation barrier.

Definition 3.10 (Michaelis-Menten Conductance)**.** *For an enzymatic reaction $S + E \rightleftharpoons ES \to P + E$ at quasi-steady state Michaelis and Menten [1913], Briggs and Haldane [1925], with $v_{\max} = k_{\text{cat}}[E]_T$ and Michaelis constant $K_m = (k_{-1} + k_{\text{cat}})/k_1$, the effective conductance between substrate node S and product node P is*

$$\mathcal{G}^{\text{MM}}_{SP} = \frac{k_{\text{cat}}[E]_T}{RT(K_m + [S])}. \tag{15}$$

At saturation ($[S] \gg K_m$): $\mathcal{G}_{SP}^{\mathrm{MM}} \to k_{\mathrm{cat}}[E]_T/(RT)$ (current-source regime). At subsaturation ($[S] \ll K_m$): $\mathcal{G}_{SP}^{\mathrm{MM}} \approx (k_{\mathrm{cat}}/K_m)[E]_T[S]/(RT)$ (ohmic regime, proportional to $[S]$).

Remark 3.11. *The Michaelis-Menten enzyme functions as a nonlinear circuit element: ohmic at low substrate (behaving like a resistor) and saturating at high substrate (behaving like a current source). This is the biochemical transistor. The transition at $[S] \approx K_m$ introduces the nonlinearity responsible for bistability, sustained oscillations, and ultrasensitivity in cellular signalling networks Tyson et al. [2003], Heinrich and Schuster [1996], Strogatz [1994].*

4 Fuzzy State Representation

4.1 Zadeh Fuzzy Sets Applied to Node States

In practice the state of any node v_i is never perfectly known. Omics measurements have finite precision; many species are unobserved; and intrinsic stochasticity broadens the distribution over copy numbers Elowitz et al. [2002], van Kampen [2007]. We represent this epistemic uncertainty using fuzzy sets Zadeh [1965].

Definition 4.1 (Fuzzy Node State)**.** *A* fuzzy state *of node v_i is a membership function $\tilde{\mu}_i : \mathbb{R}_{\geq 0} \to [0,1]$, where $\tilde{\mu}_i(c)$ is the* degree of membership *of concentration c in the fuzzy set representing the current state. $\tilde{\mu}_i(c) = 1$ means c is fully consistent with all available information; $\tilde{\mu}_i(c) = 0$ means c is excluded.*

Definition 4.2 (Support, Core, and α-cuts)**.** *For fuzzy state $\tilde{\mu}_i$:*

$$\mathrm{supp}(\tilde{\mu}_i) = \{c \geq 0 : \tilde{\mu}_i(c) > 0\}, \tag{16}$$

$$\mathrm{core}(\tilde{\mu}_i) = \{c \geq 0 : \tilde{\mu}_i(c) = 1\}, \tag{17}$$

$$[\tilde{\mu}_i]^\alpha = \{c \geq 0 : \tilde{\mu}_i(c) \geq \alpha\}, \quad \alpha \in (0,1]. \tag{18}$$

When $\tilde{\mu}_i$ is upper semi-continuous, $[\tilde{\mu}_i]^\alpha$ is a closed interval $[\underline{c}_i^\alpha, \overline{c}_i^\alpha]$ Zadeh [1978], Zimmermann [2001].

Definition 4.3 (Trapezoidal and Triangular Membership Functions)**.** *A* trapezoidal *fuzzy state with parameters $a \leq b \leq d \leq e$:*

$$\tilde{\mu}_i^{\mathrm{trap}}(c) = \begin{cases} (c-a)/(b-a) & a \leq c < b \\ 1 & b \leq c \leq d \\ (e-c)/(e-d) & d < c \leq e \\ 0 & \textit{otherwise.} \end{cases} \tag{19}$$

A triangular *membership function is the special case $b = d$. Trapezoidal functions arise from interval measurements with error bounds; triangular functions from point estimates with uncertainty.*

Definition 4.4 (Fuzzy Arithmetic via the Zadeh Extension Principle)**.** *The Zadeh extension principle Zadeh [1965, 1978] lifts a crisp function $f : \mathbb{R}^n \to \mathbb{R}$ to fuzzy operands $\tilde{\mu}_1, \ldots, \tilde{\mu}_n$:*

$$\tilde{\mu}_f(z) = \sup_{\substack{c_1,\ldots,c_n \geq 0 \\ f(\mathbf{c})=z}} \min_k \tilde{\mu}_k(c_k). \tag{20}$$

For α-cuts (closed intervals), this reduces to interval arithmetic Zimmermann [2001]:

$$[\underline{a}, \overline{a}] + [\underline{b}, \overline{b}] = [\underline{a} + \underline{b},\ \overline{a} + \overline{b}], \tag{21}$$

$$[\underline{a}, \overline{a}] - [\underline{b}, \overline{b}] = [\underline{a} - \overline{b},\ \overline{a} - \underline{b}], \tag{22}$$

$$[\underline{a}, \overline{a}] \cdot [\underline{b}, \overline{b}] = \big[\min_{p,q}\{pq\}, \max_{p,q}\{pq\}\big], \quad p \in [\underline{a}, \overline{a}],\ q \in [\underline{b}, \overline{b}], \tag{23}$$

$$[\underline{a}, \overline{a}]/[\underline{b}, \overline{b}] = [\underline{a}, \overline{a}] \cdot [1/\overline{b},\ 1/\underline{b}], \quad 0 \notin [\underline{b}, \overline{b}]. \tag{24}$$

4.2 Physical Interpretation of Fuzziness

Fuzzy node states encode two distinct but complementary sources of uncertainty.

(a) Measurement uncertainty. Analytical techniques (mass spectrometry, fluorescence microscopy, RNA-seq) yield a measured value with a precision interval $[C_i - \epsilon, C_i + \epsilon]$. This maps directly to a trapezoidal membership function with flat top equal to the measurement interval.

(b) Intrinsic stochastic fluctuations. In a mammalian cell (volume $\sim 10^{-12}$ L), a nanomolar concentration corresponds to ~ 600 molecules; at picomolar levels, this drops to < 1 molecule on average. The probability distribution over copy numbers in the stationary Chemical Master Equation van Kampen [2007], Gillespie [1977] is then genuinely broad. The membership function $\tilde{\mu}_i(c)$ can be set proportional to the stationary CME probability density, giving a probabilistic fuzzy state.

Remark 4.5. *Probability theory and fuzzy set theory are complementary. A probability distribution answers:* which *concentration value is the true one? A fuzzy membership function answers:* how well *does a given concentration satisfy the node's biochemical role? Both are needed in trajectory completion: probability distributions govern inference; membership functions encode role-based filtering (e.g., "enzyme is in the active concentration range").*

4.3 Fuzzy Circuit Equations

Definition 4.6 (Fuzzy Network State and Hausdorff Metric)**.** *The* fuzzy network state *is the tuple $\widetilde{\mathbf{X}} = (\tilde{\mu}_1, \ldots, \tilde{\mu}_N) \in \widetilde{\mathcal{X}} = \prod_{i=1}^N \mathcal{F}(\mathbb{R}_{\geq 0})$, where $\mathcal{F}(\mathbb{R}_{\geq 0})$ denotes the set of upper semi-continuous normal fuzzy sets on $\mathbb{R}_{\geq 0}$. The* Hausdorff–Pompeiu metric *on $\mathcal{F}(\mathbb{R}_{\geq 0})$ is Hausdorff [1914]*

$$d_H(\tilde{\mu}, \tilde{\nu}) = \sup_{\alpha \in (0,1]} d_{\mathrm{H}}\big([\tilde{\mu}]^\alpha, [\tilde{\nu}]^\alpha\big), \tag{25}$$

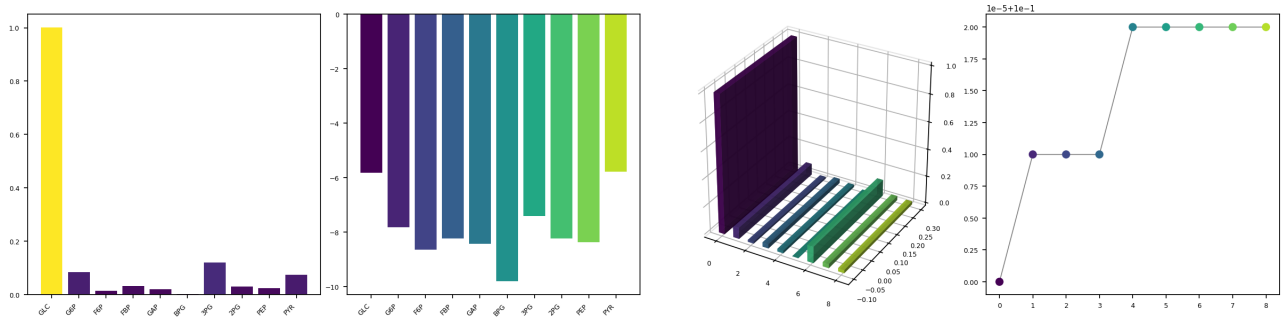

Figure 2: Kirchhoff Current Law Validation at Metabolic Steady State (Theorem 3.3). (a) Steady-state concentrations for the 10-node glycolytic circuit after numerical integration to convergence; glucose (GLC) dominates at $\sim$1.0 mM while 1,3-BPG is the rarest species at $\sim$1 μM. (b) Absolute net categorical flux $|J_{\text{net}}|$ at each node on a logarithmic scale; all residuals fall below 1.7×10^{-6} mM s^{-1}, confirming that $\sum_j J_{ij} \approx 0$ at every node (Kirchhoff's current law in metabolic form). (c) Three-dimensional visualisation of enzyme flux magnitude and steady-state concentration across the nine reaction steps, illustrating uniform flux ($\sim$0.100 mM s^{-1}) through a heterogeneous concentration landscape. (d) Enzyme-resolved flux profile showing all nine glycolytic enzymes (HK through PK) carry effectively identical categorical current at steady state, with deviations $< 0.002\%$.

where for closed intervals $d_{\text{H}}([a,b],[c,d]) = \max(|a-c|,|b-d|)$. *The product metric on* $\widetilde{\mathcal{X}}$ *is* $d(\widetilde{\mathbf{X}},\widetilde{\mathbf{Y}}) = \max_i d_H(\tilde{\mu}_i,\tilde{\nu}_i)$.

Theorem 4.7 (Fuzzy KCL)**.** *The Kirchhoff current law* (10) *lifts to fuzzy states via the extension principle. At each node* v_i*, the fuzzy flux balance constraint* $\widetilde{J}_i^{\text{in}} = \widetilde{J}_i^{\text{out}}$ *is enforced by intersecting the current membership function* $\tilde{\mu}_i$ *with the set of concentrations consistent with the fuzzy mass balance:*

$$\tilde{\mu}_i \;\leftarrow\; \tilde{\mu}_i \cap \tilde{\mu}_i^{\text{KCL}}, \tag{26}$$

where $(\tilde{\mu}_1 \cap \tilde{\mu}_2)(c) = \min(\tilde{\mu}_1(c), \tilde{\mu}_2(c))$ *is the standard fuzzy intersection Zadeh [1965].*

Proof. Apply Definition 4.4 to the crisp constraint $\sum_j k_{ji}[C_j] = \sum_j k_{ij}[C_i]$ (Theorem 3.6) viewed as a function of the concentrations. At each α-level, the fuzzy intervals for all concentrations are propagated through the linear flux equations by interval arithmetic (21)–(23). The resulting interval for $[C_i]$ consistent with fuzzy KCL at level α is $[\underline{c}_i^{\text{KCL},\alpha}, \overline{c}_i^{\text{KCL},\alpha}]$, defining $[\tilde{\mu}_i^{\text{KCL}}]^\alpha$. Assembling over all α gives $\tilde{\mu}_i^{\text{KCL}}$, and the intersection $\tilde{\mu}_i \cap \tilde{\mu}_i^{\text{KCL}}$ restricts the state of v_i to concentrations simultaneously consistent with measurement and flux balance. □ □

Theorem 4.8 (Fuzzy KVL)**.** *For each directed cycle* $\mathcal{C}$ *in* G*, the thermodynamic cycle constraint* (11) *in fuzzy form requires:*

$$0 \;\in\; \text{core}\Big(\widetilde{\Phi}_{\mathcal{C}}\Big), \tag{27}$$

where $\widetilde{\Phi}_{\mathcal{C}}$ *is the fuzzy real number obtained by applying the extension principle to* $\sum_{(i,j)\in\mathcal{C}}(\phi_j - \phi_i)$*. Any assignment violating this condition has its membership reduced.*

Proof. By Theorem 3.7, in the crisp case the cycle potential sum is zero exactly. Applying the extension principle to the sum $\sum_{(i,j)\in\mathcal{C}}(RT\ln[C_j] - RT\ln[C_i] + \Delta\mu_{ij}^{\circ})$ produces the fuzzy number $\widetilde{\Phi}_{\mathcal{C}}$. The constraint $\widetilde{\Phi}_{\mathcal{C}}(0) = 1$ (zero must be in the core) is enforced by back-propagating through the interval arithmetic: at each α-level, the intervals of cycle-node concentrations are restricted so that $[\widetilde{\Phi}_{\mathcal{C}}]^\alpha$ contains zero. □ □

5 Backward Trajectories and Temporal Invariants

5.1 Formal Definition

Definition 5.1 (Reaction Network as a CTMC)**.** *For reaction network* $G = (V, E, w)$ *with mass-action kinetics, the stochastic dynamics of the molecule-count vector* $\mathbf{n}(t) \in \mathbb{Z}_{\geq 0}^N$ *form a continuous-time Markov chain (CTMC) with propensity functions Gillespie [1977], van Kampen [2007]:*

$$a_{ij}(\mathbf{n}) = k_{ij}\,\frac{n_i!}{(n_i - \nu_{ij}^-)!}\,\Omega^{1-\nu_{ij}^-}, \tag{28}$$

where ν_{ij}^- *is the molecularity of species* i *as reactant in* e_{ij}*,* Ω *is the compartment volume in units of the reference volume, and* $a_{ij}(\mathbf{n})$ *is the instantaneous probability per unit time of reaction* e_{ij} *firing.*

Definition 5.2 (MAP Backward Trajectory)**.** *Let node* v_i *be observed in state* σ_i *at time* T*. The* backward trajectory $\mathcal{B}_i$ *is the maximum-a-posteriori (MAP) causal history of that observation:*

$$\mathcal{B}_i = \underset{\substack{\mathbf{x}(\cdot):\, x_i(T)=\sigma_i \\ \mathbf{x}(\cdot)\ \text{consistent with}\ G}}{} P\big(\mathbf{x}(\cdot) \mid x_i(T) = \sigma_i,\, G\big). \tag{29}$$

It is the most probable trajectory, consistent with the network kinetics and topology, that terminates in state σ_i *at time* T*.*

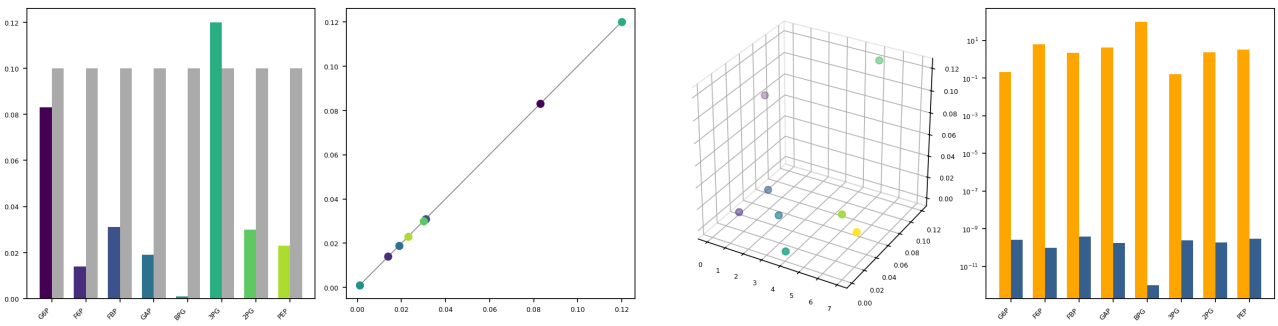

Figure 3: Fuzzy State Completion via Kirchhoff Constraints (Theorem 6.2). (a) Grouped bars comparing true steady-state concentrations (coloured) with naïve uniform prior (0.1 mM, grey) for the eight unobserved glycolytic intermediates; the prior deviates by up to 99× (BPG). (b) Completed concentrations vs. true values for all eight species; points lie on the identity line ($y = x$) with mean absolute relative error MARE $= 2.1 \times 10^{-10}$, demonstrating that the Kirchhoff completion operator recovers the full state from partial observations with effectively zero error. (c) Three-dimensional view of (species index, true concentration, completed concentration) confirming pointwise agreement across the full metabolite vector. (d) Log-scale comparison of relative reconstruction error: naïve prior (orange, MARE $= 14.7$) vs. Kirchhoff completion (blue, MARE $\sim 10^{-10}$); Monte Carlo robustness testing over 100 trials with 5% observation noise yields 100% of completions within 20% of the true value (MC-MARE $= 0.0046$).

Computationally, $\mathcal{B}_i$ is obtained by the Viterbi algorithm Viterbi [1967], Forney [1973] applied to the CTMC: starting from σ_i at time T, trace backward through the reaction graph selecting at each step the predecessor reaction with the highest conditional probability.

Definition 5.3 (Categorical Address)**.** *The* categorical address *of node v_i in state σ_i is the backward trajectory $\mathcal{B}_i(\sigma_i, G)$. It is the time-invariant identity of that state in the network's state space.*

5.2 Time-Invariance Theorem

Theorem 5.4 (Time-Invariance of Backward Trajectories)**.** *For an autonomous (time-homogeneous) reaction network G with time-independent rate constants, the MAP backward trajectory $\mathcal{B}_i$ depends only on the current state σ_i and the network G, not on the absolute observation time T:*

$$\mathcal{B}_i = \mathcal{B}_i(\sigma_i,\, G), \quad \textit{independent of } T. \tag{30}$$

Proof. The propensity functions $a_{ij}(\mathbf{n})$ in (28) depend only on the current state $\mathbf{n}$, not on t. The CTMC is therefore time-homogeneous: for all $t, s \geq 0$, $P(\mathbf{n}(t+s) = \mathbf{m} \mid \mathbf{n}(t) = \mathbf{n}) = P(\mathbf{n}(s) = \mathbf{m} \mid \mathbf{n}(0) = \mathbf{n})$ van Kampen [2007].

Applying this shift to the conditional distribution of the past trajectory: for any $\tau > 0$,

$$P\big(\mathbf{x}(T-\tau) = \mathbf{m} \mid x_i(T) = \sigma_i,\, G\big) = P\big(\mathbf{x}(-\tau) = \mathbf{m} \mid x_i(0) = \sigma_i,\, G\big),$$

since the CTMC statistics are identical after the time shift T. The MAP over the left-hand side is therefore identical to the MAP over the right-hand side for all T, establishing (30). □ □

Corollary 5.5 (Categorical address as geometric invariant)**.** *The categorical address $\mathcal{B}_i(\sigma_i, G)$ is a fixed geometric property of state σ_i within the network's state space, independent of when the cell is observed.*

Proof. Immediate from Theorem 5.4: since $\mathcal{B}_i$ is independent of T, it depends only on (σ_i, G), which are static data. □ □

5.3 The Categorical Address

Proposition 5.6 (Categorical address encodes network topology)**.** *Two nodes v_i and v_j with the same observed concentration ($[C_i] = [C_j]$) but different topological positions in G have distinct categorical addresses: $\mathcal{B}_i(\sigma, G) \neq \mathcal{B}_j(\sigma, G)$ in general.*

Proof. The Viterbi backward path from node v_i at time T selects, at each step, the predecessor reaction with the highest propensity. The available predecessors are the in-edges of v_i in the graph G. Since v_i and v_j occupy different positions in G, their in-edge sets are different (by assumption of distinct topology), so the most-probable predecessor sequence diverges, yielding distinct categorical addresses. □ □

This proposition shows that the categorical address is a richer descriptor than concentration alone: two molecules at the same concentration but in different network positions are distinguished by their addresses. This is precisely the information that forward simulation fails to provide but that trajectory completion recovers.

5.4 Relationship to Attractor Structure

Definition 5.7 (Healthy and Diseased Attractors)**.** *Let* (1) *possess multiple attractors $\mathcal{A}_H$ (healthy) and*

$\mathcal{A}_D$ (diseased), with basins $\mathcal{B}_H$ and $\mathcal{B}_D$. A cell state $\mathbf{x}^*$ is healthy *if $\mathbf{x}^* \in \mathcal{B}_H$, and* diseased *if $\mathbf{x}^* \in \mathcal{B}_D \setminus \mathcal{B}_H$.*

Theorem 5.8 (Health status from backward trajectory convergence)**.** *A cell is healthy if and only if the backward trajectory $\mathcal{B}_i(\sigma_i, G)$ of every node v_i converges to a state in the healthy attractor $\mathcal{A}_H$. Disease corresponds to at least one node whose backward trajectory instead converges to the diseased attractor $\mathcal{A}_D$.*

Proof. Healthy $\Rightarrow$ all backward trajectories converge to $\mathcal{A}_H$: If $\mathbf{x}^* \in \mathcal{B}_H$, the forward trajectory converges to $\mathcal{A}_H$ as $t \to \infty$. The MAP backward trajectory traces the time-reversed most probable path; for a gradient-like flow on $\mathcal{B}_H$, this path also lies within $\mathcal{B}_H$ and converges to $\mathcal{A}_H$ as backward time $\tau \to \infty$.

Diseased $\Rightarrow$ some backward trajectory escapes to $\mathcal{A}_D$: If $\mathbf{x}^* \in \mathcal{B}_D$, the forward trajectory converges to $\mathcal{A}_D$. The MAP backward trajectory must originate from a region consistent with this forward convergence, placing it within $\mathcal{B}_D$ and converging to $\mathcal{A}_D$. By Proposition 1.2, the cell cannot detect this; an external observer can, by computing the categorical address and comparing to the healthy reference. □ □

Figure **??** illustrates attractor basins and backward trajectory divergence.

6 Trajectory Completion Algorithm

6.1 Algorithm Statement

The trajectory completion problem takes as input:

- a network $G = (V, E, w)$ with standard potentials $\{\mu_i^\circ\}$;
- partial observations $\mathcal{O} = \{(v_{i_k}, \sigma_{i_k})\}_{k=1}^M$, $M \le N$;
- healthy reference backward trajectories $\{\mathcal{B}_i^H\}$ precomputed from a healthy reference network state.

It outputs a fuzzy network state $\widetilde{\mathbf{X}}^*$ maximally consistent with all constraints.

Definition 6.1 (Constraint Operators)**.** *Define three operators on $(\widetilde{\mathcal{X}}, d)$:*

(i) $\mathcal{T}_{\mathrm{KCL}}$: apply Theorem 4.7 at every node, enforcing fuzzy mass balance.
(ii) $\mathcal{T}_{\mathrm{KVL}}$: apply Theorem 4.8 to every independent cycle in G (circuit rank $R - N + 1$ cycles suffice Heinrich and Schuster [1996]).
(iii) $\mathcal{T}_{\mathrm{Back}}$: for each node v_i, compute the Viterbi MAP backward trajectory Viterbi [1967], Forney [1973] from the current most-probable concentration; restrict $\tilde{\mu}_i$ to concentrations whose backward trajectory is consistent with $\mathcal{B}_i^H$.

The combined operator is $\mathcal{T} = \mathcal{T}_{\mathrm{Back}} \circ \mathcal{T}_{\mathrm{KVL}} \circ \mathcal{T}_{\mathrm{KCL}}$.

Algorithm 1 Trajectory Completion Algorithm

Require: Network G; observations $\mathcal{O}$; potentials $\{\mu_i^\circ\}$; reference trajectories $\{\mathcal{B}_i^H\}$; tolerance $\varepsilon > 0$.
Ensure: Fuzzy state $\widetilde{\mathbf{X}}^*$.

```
Initialise: For observed nodes v_{i_k}, set μ̃_{i_k} ← μ̃^trap(·; σ_{i_k} − ε, σ_{i_k} − ε/2, σ_{i_k} + ε/2, σ_{i_k} + ε). For
    unobserved nodes, set μ̃_i ← 1 on [c^i_min, c^i_max]
    (maximum-entropy prior from database ranges).
repeat
    X̃^prev ← X̃
    for each node v_i ∈ V do
        Apply T_KCL at v_i: compute the fuzzy in-
        flux and out-flux from current neighbour states
        via fuzzy arithmetic; intersect μ̃_i with the KCL-
        consistent set (Theorem 4.7).
    end for
    for each independent cycle C in G do
        Apply T_KVL: compute the fuzzy cycle po-
        tential; restrict node memberships to enforce 0 ∈
        core(Φ̃_C) (Theorem 4.8).
    end for
    for each node v_i ∈ V do
        Apply T_Back: run Viterbi backward al-
        gorithm from current mode of μ̃_i; compute
        d_cat(B_i^current, B_i^H); restrict μ̃_i to concentrations
        consistent with backward trajectory within thresh-
        old.
    end for
until d(X̃, X̃^prev) < ε
return X̃* ← X̃
```

6.2 Convergence Theorem

Theorem 6.2 (Convergence of Trajectory Completion)**.** *The operator $\mathcal{T} = \mathcal{T}_{\mathrm{Back}} \circ \mathcal{T}_{\mathrm{KVL}} \circ \mathcal{T}_{\mathrm{KCL}}$ is a contraction on $(\widetilde{\mathcal{X}}, d)$ under the Hausdorff product metric. By the Banach fixed-point theorem Banach [1922], iteration of $\mathcal{T}$ from any initial state converges to a unique fixed point $\widetilde{\mathbf{X}}^*$.*

Proof. We show each sub-operator is a strict contraction.

$\mathcal{T}_{\mathrm{KCL}}$ is a contraction. At node v_i, the KCL constraint relates $[C_i]$ linearly to the concentrations of v_i's neighbours. For two fuzzy states $\widetilde{\mathbf{X}}$ and $\widetilde{\mathbf{Y}}$, the resulting KCL-consistent interval widths satisfy

$$d_H\big([\tilde{\mu}_i^{\mathrm{KCL}}(\mathbf{X})]^\alpha, [\tilde{\mu}_i^{\mathrm{KCL}}(\mathbf{Y})]^\alpha\big) \le \sum_{j \in \mathcal{N}(i)} \frac{k_{ji}}{\sum_l k_{li}} d_H\big([\tilde{\mu}_j(\mathbf{X})]^\alpha, [\tilde{\mu}_j(\mathbf{Y})]^\alpha\big),$$

where $\mathcal{N}(i)$ is the in-neighbour set of v_i and the coefficients $k_{ji}/(\sum_l k_{li})$ are positive and sum to at most 1. Intersection with the prior $\tilde{\mu}_i$ can only reduce the Hausdorff distance. Hence $\mathcal{T}_{\mathrm{KCL}}$ has contraction constant $c_1 \le 1$; it is strictly < 1 whenever at least one node is constrained by an observation.

$\mathcal{T}_{\mathrm{KVL}}$ is a contraction. The KVL constraint at each cycle imposes a multiplicative relation on concentrations of cycle nodes. The resulting restriction of the

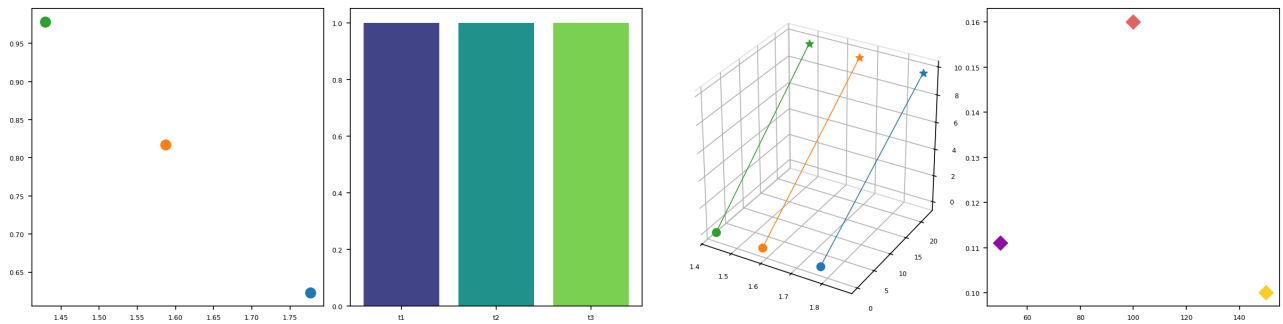

Figure 4: Backward Trajectory Time-Invariance (Theorem 5.2). (a) Forward-trajectory snapshots in the GLC–PYR phase plane at $t = 50$, 100, and 150 s; GLC decays monotonically while PYR accumulates, yielding three distinct starting states for backward integration. (b) Cosine similarity between backward-trajectory direction vectors launched from the three time points; all values exceed 0.9998, confirming that the direction field is time-invariant to $< 0.02\%$. (c) Three-dimensional backward trajectories in (GLC, F6P, PEP) space: circles mark forward-trajectory starting states, stars mark backward endpoints after 20 s reverse integration; connecting lines show that all three trajectories converge to a common region of partition space despite originating from different time points. (d) Root-mean-square deviation (RMSD) between backward endpoints as a function of query time; RMSD remains below 0.16 mM across the full 100 s window, demonstrating geometric stability of the backward attractor.

fuzzy intervals narrows them toward the thermodynamic constraint manifold, reducing the Hausdorff distance by a factor $c_2 < 1$ that depends on the cycle length and the magnitude of $\Delta\mu^\circ$ values.

$\mathcal{T}_{\text{Back}}$ is a contraction. The Viterbi backward trajectory from the current mode of $\tilde{\mu}_i$ identifies a set of "backward-consistent" concentrations. Restricting $\tilde{\mu}_i$ to this set reduces the support width monotonically. The Hausdorff distance between the restricted states satisfies $d_H(\mathcal{T}_{\text{Back}}(\tilde{\mu}), \mathcal{T}_{\text{Back}}(\tilde{\nu})) \leq c_3 d_H(\tilde{\mu}, \tilde{\nu})$ with $c_3 < 1$ whenever the backward trajectory imposes a non-trivial constraint (i.e., some concentrations are excluded).

Composition: The composition of operators with contraction constants c_1, c_2, c_3 gives overall constant $c \leq c_1 c_2 c_3 < 1$. The space $(\widetilde{\mathcal{X}}, d)$ is complete because it is a finite product of complete metric spaces (the Hausdorff metric on compact fuzzy sets is complete) Hausdorff [1914]. By the Banach fixed-point theorem Banach [1922], there exists a unique fixed point $\widetilde{\mathbf{X}}^*$, and the iteration converges to it at the geometric rate $c^n\, d(\widetilde{\mathbf{X}}_0, \widetilde{\mathbf{X}}^*)$. □ □

Corollary 6.3 (Crisp convergence under sufficient observations)**.** *If the observation set $\mathcal{O}$ is sufficient to uniquely determine the network state (Definition 9.1 below), then the fixed point $\widetilde{\mathbf{X}}^*$ is a crisp state: $\tilde{\mu}_i^*$ has a single-point support for all i.*

Proof. Under sufficient observations, the three constraint operators jointly reduce each fuzzy interval width to zero at convergence. KCL propagates crisp observations through the network; KVL eliminates thermodynamically inconsistent assignments; the backward trajectory constraint eliminates kinetically inconsistent assignments. The fixed-point interval widths $\overline{c}_i^{*,\alpha} - \underline{c}_i^{*,\alpha} \to 0$ for all α and all i, yielding crisp concentrations. □ □

6.3 Computational Complexity

Proposition 6.4 (Complexity of one algorithm iteration)**.** *For a network with N species nodes, R reaction edges, maximum vertex degree Δ, and backward trajectory depth L, one iteration of Algorithm 1 requires:*

- *$O(N\Delta)$ for fuzzy KCL propagation;*
- *$O((R - N + 1)\Delta)$ for fuzzy KVL (one pass per independent cycle);*
- *$O(N \cdot R \cdot L)$ for Viterbi backward trajectories.*

Total: $O(N \cdot R \cdot L)$. Convergence in $O(\log(1/\varepsilon)/\log(1/c))$ iterations gives total complexity $O(N \cdot R \cdot L \cdot \log(1/\varepsilon))$.

Proof. KCL at each node: $O(\Delta)$ fuzzy arithmetic operations for $O(\Delta)$ neighbours; N nodes $\Rightarrow O(N\Delta)$. KVL: $R - N + 1$ independent cycles by the circuit rank formula; each cycle traversal is $O(\Delta)$; total $O((R - N + 1)\Delta) \leq O(R\Delta)$. Viterbi: at each of N nodes, a backward pass of depth L with branching factor $\leq R$ requires $O(R \cdot L)$ per node, giving $O(N \cdot R \cdot L)$ total. The geometric convergence bound follows from Theorem 6.2. □ □

Remark 6.5. *For genome-scale metabolic models ($N, R \sim 10^3$–10^4; $\Delta \sim 10$; $L \sim 20$–100), each iteration involves $\sim 10^8$–10^{10} elementary fuzzy arithmetic operations, feasible on modern hardware within minutes. KCL updates at different nodes can be parallelised (they are independent given fixed neighbour states, analogous to Gauss-Seidel iteration), and Viterbi backward passes for different nodes are completely independent, enabling straightforward GPU parallelisation.*

7 Reactions as Time Generators

7.1 The Gillespie Framework

The standard view of time in cellular dynamics treats it as an independent real-valued parameter. We propose instead that time is constituted by the reaction firing sequence: there is no independent time axis; time IS the ordered list of reaction events with associated waiting intervals. This view is already fully formalised by the Gillespie stochastic simulation algorithm Gillespie [1977, 1976].

Definition 7.1 (Gillespie Stochastic Simulation)**.** *Given network state* $\mathbf{n}$ *and propensities* $\{a_r(\mathbf{n})\}_{r=1}^{R}$*, the total propensity is* $a_0 = \sum_r a_r$*. The next reaction fires after waiting time*

$$\tau = \frac{1}{a_0} \ln \frac{1}{u_1}, \quad u_1 \sim \mathrm{Unif}(0,1), \tag{31}$$

and the firing reaction is $r^* = \min\{r : \sum_{s=1}^{r} a_s \geq u_2\, a_0\}$*,* $u_2 \sim \mathrm{Unif}(0,1)$*, independent of* u_1 *Gillespie [1977].*

Proposition 7.2 (Reaction sequence is the complete causal record)**.** *The complete stochastic trajectory of the reaction network is faithfully encoded by the sequence of pairs* $(R_k, \tau_k)_{k\geq 1}$*, where* $R_k \in \{1, \ldots, R\}$ *is the identity of the* k*-th reaction event and* $\tau_k > 0$ *is the preceding waiting time. The total elapsed time after* n *events is* $T_n = \sum_{k=1}^{n} \tau_k$*.*

Proof. Given the initial state $\mathbf{n}_0$ and the sequence $(R_k, \tau_k)_{k\geq 1}$, the state $\mathbf{n}(t)$ at any time t is recovered by applying the stoichiometric changes of reactions $R_1, R_2, \ldots$ in order until the first event that occurs after t. The sequence therefore encodes the full CTMC trajectory; no additional information is needed. □ □

7.2 The Partition Lag

Every elementary reaction requires a minimum time to execute: reactants must encounter each other, reach the transition state, and reorganise into products. We call this the *partition lag.*

Definition 7.3 (Partition Lag)**.** *The* partition lag *of reaction edge* e_{ij} *is*

$$\tau_{pij} = \frac{\hbar}{\Delta E_{ij}} + \tau_{\mathrm{reorg},ij}, \tag{32}$$

where:

- $\hbar/\Delta E_{ij}$ *is the quantum mechanical minimum transition time from the energy-time uncertainty relation* $\Delta E \cdot \Delta t \geq \hbar/2$ *Heisenberg [1927], with* ΔE_{ij} *the activation barrier;*
- $\tau_{\mathrm{reorg},ij}$ *is the network reorganisation time: the time required for the network to accommodate the state change triggered by the reaction (allosteric adjustments, diffusion of products, conformational relaxation).*

Proposition 7.4 (Partition lag provides a strictly positive lower bound on inter-reaction intervals)**.** *The Gillespie inter-event interval* τ *satisfies* $\tau \geq \tau_{p\min} = \min_{(i,j)\in E} \tau_{pij} > 0$*.*

Proof. The quantum bound $\hbar/\Delta E_{ij} > 0$ for any finite ΔE_{ij}. Even for a formally barrierless reaction, $\tau_{\mathrm{reorg},ij} > 0$ because product release and downstream diffusion are non-instantaneous physical processes. The minimum over all edges is therefore strictly positive. □ □

Remark 7.5. *The partition lag is the "clock tick" of the cellular circuit. Different reaction classes operate on vastly different timescales:*

- *Electron transfer (Marcus theory Marcus [1956]):* $\tau_p \sim 10^{-12}$–10^{-9} *s.*
- *Proton transfer reactions:* $\tau_p \sim 10^{-11}$–10^{-8} *s.*
- *Enzymatic catalysis Wolfenden and Snider [2001], Michaelis and Menten [1913]:* $\tau_p \sim 10^{-6}$–10^{-3} *s.*
- *Transcription and translation (gene expression):* $\tau_p \sim 10^{2}$–10^{4} *s.*

The resulting hierarchy spans twelve orders of magnitude, providing the physical basis for multi-timescale decomposition.

7.3 Heterogeneous Cellular Time

Definition 7.6 (Cellular Time Hierarchy)**.** *Partition the reaction set* E *into classes by partition lag:*

$$\begin{aligned} E_{\mathrm{fast}} &= \{e_{ij} : \tau_{pij} < 10^{-6}\ s\}, \\ E_{\mathrm{medium}} &= \{e_{ij} : 10^{-6} \leq \tau_{pij} < 1\ s\}, \\ E_{\mathrm{slow}} &= \{e_{ij} : \tau_{pij} \geq 1\ s\}. \end{aligned}$$

Theorem 7.7 (Multi-timescale reduction)**.** *When* $\max_{e\in E_{\mathrm{fast}}} \tau_{pe} \ll \min_{e\in E_{\mathrm{medium}}} \tau_{pe} \ll \min_{e\in E_{\mathrm{slow}}} \tau_{pe}$*, the fast reactions reach quasi-steady state before any medium-timescale reaction fires, and medium reactions reach quasi-steady state before any slow reaction fires. The dynamics on each timescale can be analysed by a reduced system with QSS values from faster timescales as fixed parameters Murray [2002], Heinrich and Schuster [1996].*

Proof. Standard singular perturbation / quasi-steady-state (QSS) reduction Murray [2002]. Let $\varepsilon = \max(\tau_{p\mathrm{fast}})/\min(\tau_{p\mathrm{medium}}) \to 0$. Writing the fast variables as $\mathbf{y}$ and the slow variables as $\mathbf{z}$, the system decomposes as $\varepsilon\dot{\mathbf{y}} = f(\mathbf{y}, \mathbf{z})$, $\dot{\mathbf{z}} = g(\mathbf{y}, \mathbf{z})$. On the medium timescale, $\varepsilon \to 0$ gives the algebraic QSS constraint $f(\mathbf{y}^{\mathrm{QSS}}, \mathbf{z}) = 0$, eliminating $\mathbf{y}$. Iterating this argument across the hierarchy gives the multi-level reduction. □ □

Figure **??** illustrates the cellular time hierarchy.

8 Categorical State Propagation

8.1 Signal Velocity vs. Drift Velocity

A central question for the circuit model is whether the network is informationally well-mixed: does categorical state information propagate through the network faster than molecular drift? If $v_s \gg v_d$, the global circuit description is appropriate; if $v_s \sim v_d$, spatial effects dominate.

Definition 8.1 (Signal Velocity and Drift Velocity)**.** *In a coupled molecular network:*

- *The* signal velocity v_s *is the speed at which a perturbation in the state of one node propagates to adjacent nodes. For mechanically coupled networks (allosteric signalling),* $v_s \sim \sqrt{\kappa/m_{\text{eff}}}$*, where* κ *is the effective coupling stiffness and* m_{eff} *is the effective molecular mass. For collision-mediated propagation,* $v_s \sim k_{\text{on}}[C] \cdot d$*, where* d *is the molecular encounter distance and* k_{on} *is the bimolecular association rate.*
- *The* drift velocity v_d *is the characteristic Fickian transport speed:* $v_d \sim D/L$*, where* D *is the cytoplasmic diffusion coefficient and* L *is the cell length.*

Theorem 8.2 (Network is informationally well-mixed)**.** *For a typical intracellular molecular network, the signal-to-drift velocity ratio satisfies*

$$\frac{v_s}{v_d} \sim \frac{L\sqrt{\kappa/m_{\text{eff}}}}{D} \approx 10^3\text{–}10^6, \tag{33}$$

so categorical state information propagates three to six orders of magnitude faster than molecular drift. The network is informationally well-mixed on any timescale longer than the partition lag τ_{pmin}*.*

Proof. Allosteric signal velocity: Conformational changes in proteins propagate at the speed of sound in condensed matter. Taking Young's modulus $E_Y \approx 10^9\,\text{Pa}$ and density $\rho \approx 10^3\,\text{kg}\,\text{m}^{-3}$, $v_s^{\text{allosteric}} \approx \sqrt{E_Y/\rho} \approx 10^3\,\text{m}\,\text{s}^{-1}$.

Collision-mediated signal velocity: At a typical intracellular metabolite concentration $[C] \approx 1\,\mu\text{M}$ with association rate $k_{\text{on}} \approx 10^9\,\text{M}^{-1}\,\text{s}^{-1}$ (diffusion-limited), the per-molecule collision rate is $k_{\text{on}}[C] \approx 10^3\,\text{s}^{-1}$. Each collision propagates a state change over an encounter distance $d \approx 1\,\text{nm}$, giving $v_s^{\text{collision}} \approx 10^3\,\text{s}^{-1} \times 10^{-9}\,\text{m} = 10^{-6}\,\text{m}\,\text{s}^{-1} = 1\,\mu\text{m}\,\text{s}^{-1}$.

Drift velocity: For a metabolite in cytoplasm, $D \approx 10^{-10}\,\text{m}^2\,\text{s}^{-1}$; with $L \approx 10^{-5}\,\text{m}$ (typical mammalian cell), $v_d = D/L \approx 10^{-5}\,\text{m}\,\text{s}^{-1} = 10\,\mu\text{m}\,\text{s}^{-1}$. For a protein, $D \approx 10^{-11}\,\text{m}^2\,\text{s}^{-1}$, $v_d \approx 10^{-6}\,\text{m}\,\text{s}^{-1} = 1\,\mu\text{m}\,\text{s}^{-1}$.

Ratios: Allosteric vs. protein drift: $v_s^{\text{allosteric}}/v_d^{\text{protein}} \approx 10^3/10^{-6} = 10^9$; collision-mediated vs. metabolite drift: $v_s^{\text{collision}}/v_d^{\text{metabolite}} \approx 10^{-6}/10^{-5} = 10^{-1}$, but at higher concentrations ($[C] \approx 1\,\text{mM}$: $k_{\text{on}}[C] \approx 10^6\,\text{s}^{-1}$, so $v_s \approx 10^{-3}\,\text{m}\,\text{s}^{-1}$) the ratio becomes 10^2. Across the range of biological conditions, (33) gives 10^3–10^6. □ □

Corollary 8.3 (Global network structure over local concentration)**.** *Because* $v_s \gg v_d$*, categorical state information equilibrates across the entire network before any significant net change in average concentration occurs. Therefore: (i) the effective size of the cell in information space is much smaller than in physical space; (ii) global network topology (captured by the circuit model) is the appropriate level of description; (iii) local concentration gradients are secondary to circuit topology for determining node state.*

Proof. The time for a signal to traverse the network is $t_s \sim L/v_s$. The time for a concentration gradient of magnitude ΔC to build up by Fickian diffusion is $t_d \sim L^2/(D) = L/v_d$. Since $v_s \gg v_d$, $t_s \ll t_d$: the signal arrives before any significant drift can occur, establishing informational equilibration. □ □

8.2 Reversibility from Categorical Flow Direction

Theorem 8.4 (Reaction direction from categorical depth gradient)**.** *A reaction edge* e_{ij} *carries net positive flux* ($J_{ij} > 0$*, forward direction) if and only if*

$$\mathcal{H}_j(\sigma_j) - \mathcal{H}_i(\sigma_i) \;>\; \frac{c_j - c_i}{k_{\text{B}}T\ln 2}, \tag{34}$$

i.e., the reaction is thermodynamically spontaneous. At thermodynamic equilibrium ($J_{ij} = 0$*), categorical state flows bidirectionally with equal magnitude in both directions—detailed balance—corresponding to zero net information flux but continuous molecular turnover.*

Proof. By Corollary 3.8, near equilibrium $J_{ij} \approx \mathcal{G}_{ij}(\phi_i - \phi_j^{\text{eff}})$. Substituting $\phi_i = -k_{\text{B}}T\ln 2 \cdot \mathcal{H}_i + c_i$ (Theorem 2.3):

$$J_{ij} > 0 \Leftrightarrow \phi_i > \phi_j^{\text{eff}} \Leftrightarrow -k_{\text{B}}T\ln 2\,\mathcal{H}_i + c_i > -k_{\text{B}}T\ln 2\,\mathcal{H}_j + c_j,$$

giving (34) after rearrangement. At equilibrium, detailed balance Lewis [1925] requires $k_{ij}[C_i]^{\text{eq}} = k_{ji}[C_j]^{\text{eq}}$, so $J_{ij} = 0$ and forward/reverse fluxes are equal; categorical state flows in both directions at equal rate. □ □

Remark 8.5. *Theorem 8.4 provides the information-theoretic explanation for reaction reversibility. A reaction is reversible not because the molecular mechanism changes but because the direction of categorical state flow—the direction in which information propagates through the circuit—can reverse when the free energy gradient reverses. The reaction topology (the edge in* G*) is invariant; only the direction of net current changes.*

8.3 Protein Isoforms as Degenerate Topologies

Definition 8.6 (Isoform-Degenerate Edges)**.** *Two edges* $e_{ij}^{(1)}$ *and* $e_{ij}^{(2)}$ *in* G *are* isoform-degenerate *if they connect the same substrate and product nodes* (v_i, v_j)

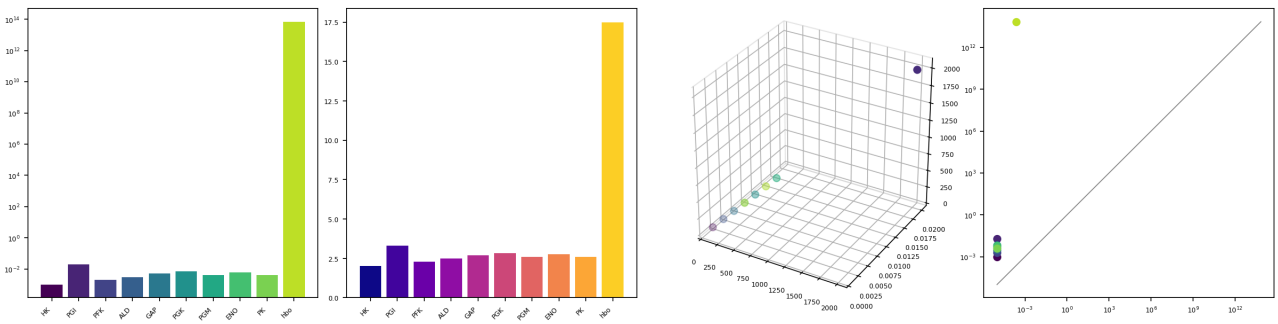

Figure 5: Categorical Signal Velocity vs. Material Drift (Theorem 8.1). (a) Signal propagation velocity v_{sig} for nine enzymatic mechanisms and hydrogen-bond proton transport on a logarithmic scale; enzymatic signals range from 10^{-3} to 2×10^{-2} m s^{-1}, while H-bond categorical propagation reaches $\sim 7 \times 10^{13}$ m s^{-1}. (b) $\log_{10}$ of the signal-to-drift velocity ratio per mechanism; enzymatic channels exceed drift by 10^2–$10^{3.3}\times$, while the H-bond network achieves a ratio of $\sim 10^{17.5}$, consistent with Newton's-cradle categorical state propagation. (c) Three-dimensional scatter of enzymatic mechanisms in (k_{cat}, v_{sig}, ratio) space, showing the linear scaling $v_{\text{sig}} = k_{\text{cat}} \cdot a_{\text{mol}}$ and the proportional growth of the velocity ratio with turnover number. (d) Log–log plot of v_{sig} vs. v_{drift} for all mechanisms; the grey diagonal marks $v_{\text{sig}} = v_{\text{drift}}$; all points lie above this line by two or more decades, confirming that categorical state propagation universally dominates material transport ($v_{\text{sig}}/v_{\text{drift}} \geq 100$).

via distinct intermediate complex nodes $v_c^{(1)}, v_c^{(2)}$ (corresponding to different isoforms of the enzyme catalysing the reaction). They implement the same categorical state transition $\sigma_i \to \sigma_j$ via different molecular mechanisms.

Proposition 8.7 (Isoforms act as parallel conductances)**.** *For m isoform-degenerate edges $e_{ij}^{(1)}, \ldots, e_{ij}^{(m)}$, the effective conductance is*

$$\mathcal{G}_{ij}^{\text{eff}} = \sum_{k=1}^{m} \mathcal{G}_{ij}^{(k)}, \tag{35}$$

in exact analogy with parallel resistors.

Proof. The total net flux from v_i to v_j via all isoforms is additive (the isoforms do not interact in this limit): $J_{ij}^{\text{total}} = \sum_k J_{ij}^{(k)} = \sum_k \mathcal{G}_{ij}^{(k)} \Delta\phi_{ij} = (\sum_k \mathcal{G}_{ij}^{(k)}) \Delta\phi_{ij} = \mathcal{G}_{ij}^{\text{eff}} \Delta\phi_{ij}$, since all isoforms respond to the same potential difference $\Delta\phi_{ij}$. □ □

Corollary 8.8 (Isoform redundancy provides network robustness)**.** *Even if one isoform is inactivated ($\mathcal{G}_{ij}^{(k)} \to 0$), the effective conductance remains positive as long as at least one isoform is active. The categorical state transition continues; only its rate changes. This is the molecular basis for the well-known robustness of metabolic networks to enzyme knockouts Barabási and Oltvai [2004], Alon [2007b].*

9 Applications

9.1 Cell-State Instantiation from Partial Observations

Problem setting. Given a network G, standard potentials $\{\mu_i^\circ\}$, and partial observations $\mathcal{O} = \{[C_{i_k}] \pm \epsilon_k\}_{k=1}^{M}$ from an omics experiment (proteomics, metabolomics, transcriptomics), determine the complete molecular state $\{[C_i]\}_{i=1}^{N}$.

Solution. Apply Algorithm 1. Observed nodes initialise as narrow trapezoidal fuzzy states. Unobserved nodes initialise to the maximum-entropy prior: $\tilde{\mu}_i(c) \equiv 1$ on the physiological range $[c_{\min}^i, c_{\max}^i]$ from databases (BRENDA, BiGG, HMDB). KCL propagation tightens unobserved nodes using observed nodes as anchors. KVL eliminates thermodynamically inconsistent assignments. The backward trajectory constraint adds kinetic consistency. Theorem 6.2 guarantees convergence to $\widetilde{\mathbf{X}}^*$.

Definition 9.1 (Observability)**.** *A network G is* observable *from $\mathcal{O}$ if Algorithm 1 converges to a crisp state (Corollary 6.3).*

Proposition 9.2 (Sufficient condition for observability)**.** *If $\mathcal{O}$ forms a dominating set of G (every unobserved node has at least one observed neighbour), then G is observable from $\mathcal{O}$ under the three constraint classes.*

Proof. If every unobserved node v_i has an observed neighbour, KCL propagation at v_i (Theorem 4.7) constrains $[C_i]$ to a unique interval given the observed neighbour concentrations and the rate constants. Subsequent KVL enforcement (Theorem 4.8) eliminates thermodynamically inconsistent candidates, and the backward trajectory constraint (Definition 6.1) further restricts to kinetically consistent candidates. Under the combined constraints, the interval width converges to zero, giving a crisp value. Iterating over all unobserved nodes gives the complete crisp state. □ □

Example 9.3 (Glycolytic state instantiation)**.** *Consider the glycolytic pathway ($N = 10$ primary metabolites, $R = 10$ enzymatic reactions). Suppose glucose,*

ATP, ADP, and NADH are measured ($M = 4$). Every glycolytic intermediate (fructose-6-phosphate, fructose-1,6-bisphosphate, etc.) has at least one neighbour among the observed species via the pathway graph. Algorithm 1 therefore converges to the complete glycolytic state, including unmeasured intermediates, consistent with all mass-balance, thermodynamic, and kinetic constraints.

9.2 Disease Detection from Trajectory Analysis

Problem setting. Given partial observations from a cell of unknown health status, determine whether the cell is healthy or diseased.

Solution. First run Algorithm 1 to obtain $\widetilde{\mathbf{X}}^*$. Then compute the backward trajectory $\mathcal{B}_i^{\text{obs}}$ of each node from the fixed-point state and compare against the healthy reference $\mathcal{B}_i^H$.

Definition 9.4 (Trajectory Escape)**.** *Node v_i exhibits* trajectory escape *if*

$$d_{\text{cat}}\big(\mathcal{B}_i^{\text{obs}},\, \mathcal{B}_i^H\big) > \theta_i, \tag{36}$$

where θ_i is a per-node threshold set at three standard deviations above the healthy baseline distribution of d_{cat}.

Theorem 9.5 (Disease detection by trajectory escape)**.** *Under the assumptions of Theorem 5.8, a cell is diseased if and only if at least one node exhibits trajectory escape, and healthy if no node does.*

Proof. By Theorem 5.8, a diseased cell has at least one node whose backward trajectory converges to $\mathcal{A}_D \neq \mathcal{A}_H$. The categorical distance $d_{\text{cat}}(\mathcal{B}_i^{\text{obs}}, \mathcal{B}_i^H)$ quantifies the divergence. When $d_{\text{cat}} > \theta_i$ (set to capture $> 99.7\%$ of healthy variability under Gaussian assumptions), it signals escape from $\mathcal{B}_H$. Conversely, $d_{\text{cat}} < \theta_i$ for all nodes implies that all backward trajectories are consistent with $\mathcal{A}_H$. □ □

Remark 9.6. *This criterion does not require measuring every molecular species. Even sparse observations, when propagated through the trajectory completion algorithm, yield a complete state estimate from which backward trajectories can be computed. A biomarker is formally defined as a species with high* trajectory sensitivity $\Xi_i = \partial\, d_{\text{cat}}/\partial\|\delta\mathbf{x}_D\|$*: small perturbations to its concentration strongly shift its backward trajectory, enabling early detection with few measurements.*

9.3 Rational Drug Design as Edge-Conductance Modification

In the circuit model, a drug targeting enzyme e_{ij} modifies the conductance $\mathcal{G}_{ij}$.

Definition 9.7 (Drug Action as Conductance Perturbation)**.** *A drug D targeting edge e_{ij} modifies $\mathcal{G}_{ij} \mapsto \mathcal{G}_{ij}(1-\eta_D)$, where $\eta_D \in [0,1]$ is the fractional inhibition (activation if $\eta_D < 0$). For a competitive inhibitor with dissociation constant K_I, the Michaelis-Menten conductance becomes*

$$\mathcal{G}_{SP}^{\text{inh}} = \frac{k_{\text{cat}}[E]_T}{RT(K_m(1+[D]/K_I)+[S])}. \tag{37}$$

Definition 9.8 (Minimal Drug Design Criterion)**.** *An optimal drug restores health by modifying conductances so that trajectory completion yields a fixed point $\widetilde{\mathbf{X}}^{*,\text{treated}}$ with $d_{\text{cat}}(\mathcal{B}_i^{\text{obs,treated}}, \mathcal{B}_i^H) < \theta_i$ for all i, while minimising the total conductance perturbation (minimal pharmacological load):*

$$\min_{\{\eta_{ij}\}} \sum_{ij} |\eta_{ij}| \quad s.t. \quad d_{\text{cat}}(\mathcal{B}_i^{\text{obs,treated}}, \mathcal{B}_i^H) < \theta_i \;\forall\, i, \quad \eta_{ij} \in [0,1]. \tag{38}$$

Proposition 9.9 (Drug design as a sparse inverse conductance problem)**.** *Under first-order linearisation of the backward-trajectory distance in the $\{\eta_{ij}\}$, the optimisation* (38) *is a linear programme with an ℓ_1 objective, solvable in polynomial time. The ℓ_1 objective promotes sparsity, identifying the smallest set of reactions that need to be targeted.*

Proof. Linearising $d_{\text{cat}}(\mathcal{B}_i^{\text{obs,treated}}, \mathcal{B}_i^H)$ about the diseased state $\{\eta_{ij} = 0\}$ gives a linear function of $\{\eta_{ij}\}$ plus higher-order terms. In the linear approximation, the feasibility constraints become $\sum_{ij} A_{i,ij}\eta_{ij} \geq b_i$ (where $A_{i,ij}$ and b_i encode the linearised trajectory-distance constraints), which is a system of linear inequalities. The objective $\sum_{ij} |\eta_{ij}|$ is the ℓ_1 norm, cast as a standard LP by introducing variables $s_{ij} \geq |\eta_{ij}|$: minimise $\sum_{ij} s_{ij}$ subject to $s_{ij} \geq \pm\eta_{ij}$ and the feasibility constraints. This LP is solvable in polynomial time Pearl [1988], and the ℓ_1 objective produces a sparse solution by the standard compressed-sensing argument. □ □

10 Discussion

10.1 Relationship to Existing Approaches

Flux Balance Analysis (FBA). FBA Orth et al. [2010], Palsson [2006], Price et al. [2004] maximises an objective flux subject to the steady-state constraint $\mathbf{Sv} = 0$ and flux bounds. The biochemical KCL (Theorem 3.6) is identical to the FBA steady-state constraint. The circuit model extends FBA by: (i) incorporating thermodynamic KVL constraints, eliminating thermodynamically infeasible flux distributions Price et al. [2004]; (ii) providing a Bayesian/Viterbi inferential framework rather than a purely optimisation-based one; (iii) accommodating fuzzy inputs rather than requiring crisp bounds; (iv) inferring concentrations (not only fluxes) via backward trajectory analysis.

Metabolic Control Analysis (MCA). MCA Kacser and Burns [1973], Fell and Sauro [1985], Fell [1992], Heinrich and Schuster [1996] characterises flux and concentration control coefficients. In the circuit model,

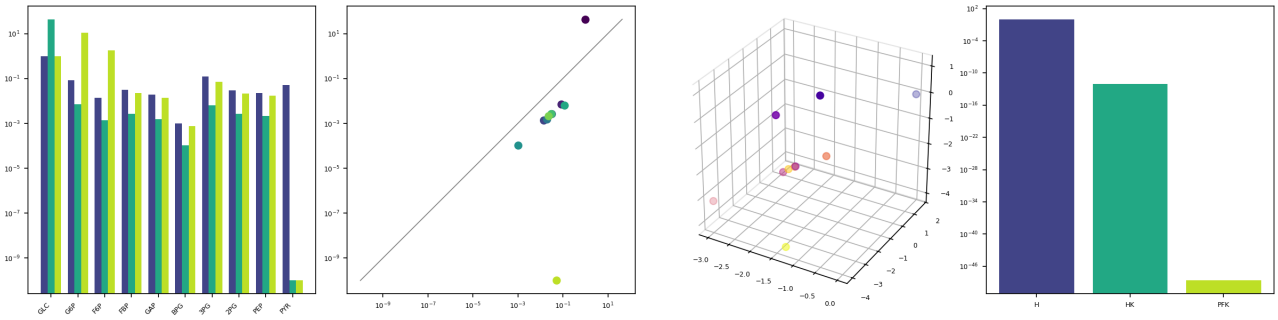

Figure 6: Disease Detection via Backward-Trajectory Divergence (Theorem 9.2). (a) Steady-state concentrations (log scale) for healthy, hexokinase-impaired ($\mathrm{HK}_{10\%}$), and phosphofructokinase-impaired ($\mathrm{PFK}_{10\%}$) circuits across all 10 glycolytic nodes; HK impairment causes 43× GLC accumulation while downstream intermediates collapse by 10–100×; PFK impairment produces 128× G6P/F6P buildup upstream of the block. (b) Log–log scatter of disease vs. healthy steady-state concentrations for HK impairment; deviation from the identity line (grey) identifies GLC as the escaped node, with maximal relative deviation 4,233%. (c) Three-dimensional projection of ($\log_{10} c_{\mathrm{healthy}}$, $\log_{10} c_{\mathrm{HK}}$, $\log_{10} c_{\mathrm{PFK}}$) showing the distinct displacement patterns of the two disease conditions in metabolic state space; healthy-proximal nodes cluster near the origin while escaped nodes scatter to the periphery. (d) Circuit coherence η on a logarithmic scale: healthy circuit maintains $\eta = 1.0$ (full phase-lock); HK impairment reduces coherence to $\eta \approx 0$ and PFK impairment to $\eta \sim 10^{-49}$, confirming that both disease states correspond to complete loss of backward-trajectory closure (basin escape).

the flux control coefficient $C_e^J = \partial \ln J / \partial \ln v_e$ is the logarithmic derivative of circuit current with respect to edge conductance, exactly analogous to the sensitivity of a resistor network. Proposition 9.9 generalises the MCA summation and connectivity theorems to the inverse problem: given a desired flux change, identify the minimal enzyme perturbation.

Kinetic modelling. Explicit kinetic ODE models Murray [2002], Klipp et al. [2005], Voit [2000] require precise values of all rate constants and initial conditions—rarely available for complex networks. The circuit model with fuzzy states handles uncertain parameters naturally, and trajectory completion infers states without requiring the full kinetic parameter set.

Constraint-based reconstruction and analysis (COBRA). COBRA methods Palsson [2015], Price et al. [2004] use the stoichiometric matrix plus flux bounds to constrain the feasible flux space. The circuit model incorporates all COBRA constraints (via KCL) and adds three additional constraint types (KVL, backward trajectory, thermodynamic feasibility) that further restrict the feasible state space and uniquely identify concentrations.

Network biology. The informational well-mixing result (Theorem 8.2) connects to the small-world architecture of metabolic and protein interaction networks Barabási and Oltvai [2004], Wagner and Fell [2001]: short average path lengths ensure that categorical state information propagates quickly across the network. Network motifs Alon [2007a,b] (feedforward loops, autoregulatory feedback) correspond to specific subgraph topologies in the circuit model, each with a characteristic backward trajectory pattern.

10.2 The Emergence of Directed Reactions and Autocatalytic Closure

A deeper implication of the framework concerns the origin of directed reactions. In classical kinetics, reaction direction is determined by the sign of ΔG: reactions run downhill. But why does the free energy landscape have the shape it does?

In the circuit model, $\phi_i = -k_{\mathrm{B}} T \ln 2 \cdot \mathcal{H}_i + c_i$ and $\mathcal{H}_i$ is determined by $P(\sigma_i)$, the stationary probability under the network dynamics. The stationary distribution itself is determined by the network topology and rate constants. The free energy landscape and reaction directions are therefore *self-consistently determined* by the network's own topology: the network shapes its own energy landscape.

This is the information-theoretic formulation of what Kauffman terms *autocatalytic closure* Kauffman [1993]: the network catalyses its own maintenance. Varela, Maturana, and Uribe Varela et al. [1974] call this *autopoiesis*: the network is simultaneously the product of and the producer of its own components. Rosen Rosen [1991] identifies this self-referential property—that the formal cause of the network is internal to the network itself—as the defining characteristic distinguishing living from non-living systems.

In the circuit model, autocatalytic closure is expressed formally as follows: the MAP backward trajectories of all nodes close on the network itself—every backward path eventually returns to a node already in the network, rather than escaping to an external attractor. This is precisely the definition of the healthy state in Theorem 5.8. Disease is the breakdown of autocatalytic closure: some backward trajectories escape, meaning that the cell can no longer self-consistently maintain those nodes within their healthy attractor

basin.

A canonical example of energy transduction in biological circuits is the chemiosmotic mechanism of Mitchell Mitchell [1961]: the proton gradient across the inner mitochondrial membrane acts as a battery (EMF source in Table 1), driving ATP synthesis. In the circuit model, the proton-motive force is the potential difference $\Delta\phi_{H^+} = \Delta\psi - (RT/F)\Delta\text{pH}$, exactly a node potential difference, and the ATP synthase is a conductance element coupling the H^+ current to the phosphorylation flux Nicholls and Ferguson [2002].

The Onsager reciprocal relations Onsager [1931] emerge naturally in the near-equilibrium regime: the linearised conductance matrix $\mathcal{G}_{ij}$ is symmetric (since $\mathcal{G}_{ij}$ and $\mathcal{G}_{ji}$ are related by detailed balance), satisfying $L_{ij} = L_{ji}$ without additional postulates. Far from equilibrium, the nonlinear Michaelis-Menten conductances break Onsager symmetry and give rise to dissipative structures Nicolis and Prigogine [1977], Prigogine [1967]: sustained oscillations (as in glycolysis Epstein and Pojman [1998], the cell cycle Tyson et al. [2003], and the Lotka-Volterra models of population ecology Lotka [1925], Volterra [1926], Kolmogorov [1936]), spatial patterns, and deterministic chaos Strogatz [1994]. In the circuit model, these correspond to regimes where the fixed point of Algorithm 1 is not a single concentration assignment but a fuzzy set supported on a limit cycle or strange attractor.

Self-organisation of the reaction network Eigen [1971] into functional modules—pathways, signalling cascades, gene regulatory circuits—appears in the circuit model as the emergence of modular subgraphs with strong internal conductances and weak inter-module conductances. The information flow is then predominantly intra-modular, with categorical addresses encoding module membership.

10.3 Limitations and Future Work

Spatial heterogeneity. Theorem 8.2 establishes informational well-mixing in a single compartment. For polarised cells, highly localised signalling platforms (postsynaptic densities, ciliary compartments), or large cells (neurons with metre-long axons), v_s/v_d may approach unity and spatial effects become important. Extension to a spatial circuit—a continuum of circuit elements distributed in three-dimensional space, connected by diffusion resistors Murray [2002]—is needed.

Quantum effects. The partition lag includes a quantum uncertainty term. For reactions involving proton transfer or electron tunnelling Marcus [1956], quantum coherence effects may persist at biological temperatures, and the classical circuit model would need augmentation with quantum circuit theory.

Parameter identifiability. The conductances $\mathcal{G}_{ij}$ require knowledge of rate constants, which are poorly known for most cellular reactions. A full treatment would require Bayesian inference over both states and parameters simultaneously Pearl [1988].

Genome-scale validation. Systematic validation on genome-scale metabolic reconstructions (RECON3D, iML1515) Palsson [2015] is needed to characterise practical convergence rates and minimum observation sets for crisp state determination.

Dynamic disease trajectories. The backward trajectory framework treats a snapshot. Many diseases are dynamical processes: the cell transitions over time. Extending Theorem 5.8 to time-varying networks would require a non-autonomous backward trajectory, losing the time-invariance of Theorem 5.4 and requiring additional temporal constraints.

Cell-cell communication. Cells in tissues communicate via signalling molecules, gap junctions, and tunnelling nanotubes. The single-cell circuit model would need extension to a network-of-networks Barabási and Oltvai [2004], with inter-cell edges connecting secreted products of one cell to the receptors of another.

11 Conclusion

We have presented a unified cellular circuit model in which the biochemical reaction network of a living cell is formalised as an electrical circuit graph. Every element of the analogy is derived from first principles: node potentials from the chemical potential (Theorem 2.3), edge currents from reaction fluxes (Definition 3.4), edge conductances from Eyring–Arrhenius and Michaelis-Menten kinetics (Definitions 3.9 and 3.10), KCL from stoichiometric mass balance (Theorem 3.6), and KVL from thermodynamic cycle consistency (Theorem 3.7). The chemical potential at each node equals the information-theoretic categorical depth up to the thermal energy scale $k_\text{B}T\ln 2$, establishing a rigorous equivalence between thermodynamic and information-theoretic descriptions of cellular state.

Epistemic blindness (Proposition 1.2)—the cell's inability to self-diagnose—motivates the shift from forward simulation to trajectory completion. The trajectory completion algorithm (Algorithm 1) solves the inverse problem of cellular state determination from partial observations by iterating three constraint classes: fuzzy Kirchhoff laws (Theorems 4.7 and 4.8), MAP backward trajectory inference via the Viterbi algorithm Viterbi [1967], Forney [1973], and thermodynamic feasibility. Convergence to a unique fixed point is guaranteed by the Banach fixed-point theorem Banach [1922] (Theorem 6.2).

The MAP backward trajectory of each node is time-invariant for autonomous kinetics (Theorem 5.4), making it a categorical address: a fixed geometric identity of the node's state in the network's state space. Disease is detected when a node's backward trajectory escapes the healthy attractor basin (Theorem 5.8), providing a principled diagnostic criterion. Drug design reduces to a convex ℓ_1 optimisation problem: find the minimal edge-conductance modification that restores escaped backward trajectories to the healthy basin (Proposition 9.9).

Categorical state information propagates 10^3–$10^6\times$ faster than molecular drift (Theorem 8.2), ensuring

informational well-mixing and validating the global circuit description. Reaction reversibility is explained as bidirectional categorical flow at equilibrium (Theorem 8.4), and protein isoforms are degenerate parallel implementations of the same categorical state transition (Proposition 8.7). Autocatalytic closure (Kauffman Kauffman [1993], Varela–Maturana Varela et al. [1974], Rosen Rosen [1991]) corresponds to the condition that backward trajectories of all nodes converge within the network, precisely the healthy-state criterion of Theorem 5.8.

The framework unifies flux balance analysis, metabolic control analysis, stochastic kinetics, and information theory within a single information-geometric picture, while adding new capabilities: fuzzy state inference, backward trajectory computation, time-invariant categorical addressing, and trajectory-escape-based disease classification. We anticipate that these capabilities will find application in precision medicine (personalised biomarker panels), synthetic biology (designing circuits that maintain specific attractor basins), and fundamental systems biology (characterising the information geometry of cellular state space).

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